Indira College of Engineering & Management, Pune

(An Autonomous Institute affiliated to SPPU)

Department of MCA

Research Project Proposal

(SY MCA – Semester III)

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1. Title of the Research Project

AI-Powered Cyber Defense and Security Analytics Platform for Intelligent Threat Detection and Risk Assessment

2. Student Details

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- Guide Name: _____
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4. Introduction

Cybersecurity threats have become increasingly sophisticated, frequent, and difficult to detect using conventional security mechanisms. Traditional signature-based security systems primarily depend on previously identified attack patterns, making them less effective against previously unseen, modified, or rapidly evolving attacks. Machine Learning (ML) and Artificial Intelligence (AI) provide an opportunity to analyze large volumes of network and security data and identify suspicious patterns based on learned behavioral characteristics.

Intrusion Detection Systems (IDS) are an important component of modern cyber defense. Publicly available datasets such as CIC-IDS2017 and UNSW-NB15 provide labeled network traffic containing both legitimate activities and different categories of attacks, enabling researchers to develop and evaluate machine-learning-based intrusion detection approaches. CIC-IDS2017 includes labeled network flows and packet captures representing common attacks, while UNSW-NB15 contains modern normal activities and multiple contemporary attack categories.

However, detecting an attack is only one part of cybersecurity operations. Security analysts also need to understand the severity of an event, identify its characteristics, prioritize alerts, and investigate potentially malicious behavior. Therefore, the proposed research focuses on developing an AI-powered cyber defense and security analytics platform capable of analyzing network/security data, detecting suspicious activity using machine-learning techniques, assigning risk levels, and presenting security information through an interactive monitoring interface.

The proposed system will investigate the effectiveness of machine-learning-based threat detection and security analytics in improving the identification and prioritization of malicious network activity. The research will also explore how detected activities can be mapped to recognized adversarial behaviors using frameworks such as MITRE ATT&CK, which provides a standardized knowledge base for understanding attacker tactics and techniques.

5. Problem Statement

Conventional cybersecurity systems often rely heavily on predefined signatures, rules, and known indicators of compromise. Such approaches can be effective against known threats but may have limitations when dealing with evolving attack patterns, previously unseen behavior, large volumes of network traffic, and false-positive alerts.

Security analysts also face the challenge of processing large quantities of security events and determining which events require immediate attention. A system that only identifies whether traffic is malicious may not provide sufficient information for effective security analysis and response.

Machine Learning can identify statistical and behavioral patterns within network traffic and has therefore been investigated as a promising approach for intrusion and anomaly detection. However, existing ML-based approaches can face challenges including class imbalance, false positives, variations between datasets and real-world traffic, limited interpretability, and difficulty detecting previously unseen attacks. Research on ML-based zero-day detection has highlighted these challenges, particularly around accuracy, recall, and generalization across attack types.

Therefore, there is a need for an integrated security analytics platform that combines machine-learning-based threat detection with risk assessment, visualization, explainable security information, and threat-oriented analysis. The proposed research aims to address this problem by designing and evaluating an AI-powered cyber defense platform capable of identifying suspicious activity and providing meaningful security insights to assist analysts in prioritizing potential threats.

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6. Aim of the Study

To design, develop, and evaluate an AI-powered cyber defense and security analytics platform that uses machine learning to detect suspicious network activity, classify potential threats, assess their risk level, and present actionable security insights to support cybersecurity monitoring and analysis.

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7. Objectives of the Study

- Objective 1:
  - To study existing AI/ML-based intrusion detection and security analytics techniques and identify their limitations in threat detection, false-positive reduction, and generalization.
- Objective 2:
  - To preprocess and analyze network/security datasets and identify relevant features that can be used for effective threat detection.
- Objective 3:
  - To develop and compare suitable machine-learning models for detecting and classifying malicious network activities.
- Objective 4:
  - To design a risk assessment mechanism that prioritizes detected security events according to their predicted threat characteristics and severity.
- Objective 5:
  - To develop an interactive security analytics dashboard for real-time or near-real-time visualization of detected threats, alerts, statistics, and security events.
- Objective 6:
  - To evaluate the proposed system using standard machine-learning and cybersecurity metrics such as accuracy, precision, recall, F1-score, false-positive rate, detection rate, and processing time.

8. Literature Review (Summary)

Machine Learning-based Intrusion Detection Systems have been extensively researched as an alternative or complement to traditional signature-based security mechanisms. ML techniques can learn patterns from network traffic and classify activities as normal or malicious. However, the effectiveness of these systems depends strongly on the quality, diversity, and representativeness of the training data.

CIC-IDS2017 was developed to provide a more realistic intrusion-detection dataset containing benign traffic and several contemporary attack scenarios. The dataset provides packet captures and labelled network flows that can be used for machine-learning-based security research.

UNSW-NB15 was created using a combination of real modern normal activities and synthetic contemporary attack behaviours. It contains nine attack categories and 49 generated features, making it another important benchmark for network intrusion detection research.

Research has also demonstrated that deep-learning-based intrusion detection systems may themselves be vulnerable to adversarial manipulation, indicating that high classification accuracy alone does not guarantee robustness in security-critical environments.

Furthermore, research into machine-learning-based zero-day attack detection identifies challenges related to generalization, accuracy, recall, and detecting previously unseen attacks.

From an industry perspective, threat-informed security frameworks such as MITRE ATT&CK are widely used to describe adversary tactics and techniques and to develop detection analytics. MITRE specifically provides guidance for using ATT&CK to develop behavioural detection and analytics capabilities.

Research Gap:
Existing research frequently focuses on improving the classification performance of individual intrusion detection models. However, there remains a practical need for integrated security analytics systems that combine ML-based detection with threat prioritization, risk assessment, visualization, interpretability, and threat-informed analysis. The proposed research addresses this gap by integrating these capabilities into a single AI-powered cyber defense prototype and evaluating its effectiveness using measurable cybersecurity and machine-learning metrics.

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9. Research Questions / Hypothesis

Research Questions

RQ1

How effectively can machine-learning techniques detect and classify malicious network activities?

RQ2

Which selected machine-learning model provides the best balance between detection performance and false-positive rate for the selected dataset?

RQ3

How effectively can risk scoring improve the prioritization of detected security events?

RQ4

Can an integrated security analytics dashboard improve the interpretability and usability of ML-based threat detection results?

RQ5

How well does the proposed detection approach generalize across different categories of network attacks?

Hypothesis

H_1 — Alternative Hypothesis

An AI/ML-based security analytics approach can effectively detect and classify malicious network activities while providing useful risk prioritization compared with a conventional rule/signature-based approach.

H_0 — Null Hypothesis

An AI/ML-based security analytics approach does not provide a statistically significant improvement in threat detection and risk prioritization compared with the selected baseline approach.

10. Scope of the Study

The proposed study focuses on the development and evaluation of an AI-powered cybersecurity analytics platform for network threat detection.

The system will include network/security data ingestion, preprocessing, feature extraction, machine-learning-based threat detection, attack classification, risk scoring, alert generation, security-event visualization, and analytical reporting.

The research will primarily use publicly available cybersecurity datasets such as CIC-IDS2017 and/or UNSW-NB15 for model development and evaluation. CIC-IDS2017 contains labeled network flows representing benign activity and multiple attack types, while UNSW-NB15 provides a separate benchmark containing modern normal and malicious traffic.

The system may additionally support controlled laboratory traffic or simulated security events for demonstration and validation purposes.

The study will focus on defensive cybersecurity activities such as detection, classification, monitoring, analysis, and alert prioritization. It will not attempt to perform unauthorized attacks against external systems or replace enterprise Security Information and Event Management (SIEM), Endpoint Detection and Response (EDR), or Security Operations Center (SOC) platforms.

The research will be limited by dataset quality, dataset-to-real-world differences, computational resources, available training data, model generalization, and the scope of the academic prototype.

11. Research Methodology

- Type of Research: Quantitative and Experimental Research
- Data Collection Method: Secondary Data (CIC-IDS2017 and UNSW-NB15) + Experimental Data
- Sample Size: Dataset-based research: 100,000+ network-flow records selected from publicly available cybersecurity dataset
- Sampling Technique: Stratified Sampling / Dataset-Based Sampling
- Tools / Technology to be Used: FastAPI, React.js, PostgreSQL, JWT, bcrypt, Pydantic, Wireshark, MITRE ATT&CK
- Data Analysis Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, and Microsoft Excel

12. Tentative Work Plan / Timeline

Phase / Activity	Duration	Remarks
Literature Review	2 Weeks	Study ML-based IDS, anomaly detection, cybersecurity analytics and MITRE ATT&CK
Requirement Analysis / Design	2 Weeks	Define architecture, data pipeline, ML workflow and dashboard
Dataset Preparation	2 Weeks	Dataset collection, cleaning, preprocessing and feature selection
ML Model Development	3 Weeks	Train and compare selected ML algorithms
Platform Implementation	4 Weeks	Backend, dashboard, alerts, risk scoring and analytics
Testing & Evaluation	2 Weeks	Model evaluation, system testing and performance analysis
Documentation & Report Writing	2 Weeks	Research report, results and final presentation

13. Expected Outcomes

The proposed research is expected to produce a functional AI-powered cyber defense and security analytics platform capable of detecting and classifying suspicious network activities using machine-learning techniques.

The study is expected to identify the most suitable machine-learning model among the selected algorithms based on evaluation metrics such as precision, recall, F1-score, false-positive rate, and detection rate.

The proposed platform is also expected to provide risk-based prioritization of detected security events and present security information through an interactive dashboard, thereby improving the interpretability and usability of ML-based detection results.

The research is expected to demonstrate the advantages and limitations of applying machine learning to cybersecurity threat detection and provide experimental findings regarding model performance across different categories of network attacks.

The project may serve as a foundation for future development of more advanced Security Operations Center (SOC), intrusion

14. References

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Y. Guo, "A Review of Machine Learning-Based Zero-Day Attack Detection: Challenges and Future Directions," *Computer Communications*, vol. 198, 2023.

4. Deep Learning IDS

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B. Strom et al., "Finding Cyber Threats with ATT&CK-Based Analytics," MITRE, 2017.

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8. Adversarial Machine Learning

A. Vassilev, A. Oprea, A. Fordyce, and H. Anderson, "Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations," NIST AI 100-2 E2023, 2024.

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10. Automated Adversary Emulation

MITRE, "CALDERA," MITRE Corporation, 2025.

Declaration by Student

I hereby declare that the proposed research work is original and will be carried out under the guidance of the assigned project guide.

Student Signature: _____

Date: _____

Guide's Recommendation

The proposed research project proposal has been reviewed and is recommended for approval.

Guide's Signature: _____

Name: _____

Date: _____